The Middle Income Trap

What separates countries that break through from those that stall?

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Who is this for, and why does it matter?

Audience: Policymakers and development researchers in middle-income countries.

Background: Some countries hit middle-income status and then get stuck there for decades. Others power through to high income. We wanted to figure out what actually makes the difference.

Setup:

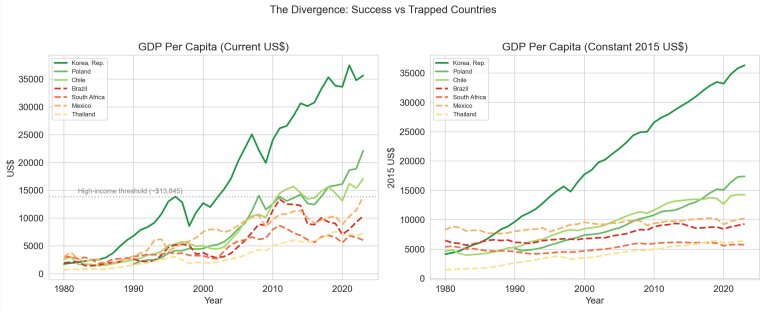
- **Success group:** South Korea, Poland, Chile — broke through
- **Trapped group:** Brazil, South Africa, Mexico, Thailand — stalled
- 15 indicators from the World Bank API, 1980–2023
- Two additional datasets: governance quality and R&D / innovation

Bottom line: what did we find?

1. The GDP gap was small in the 1980s ($1,500) but exploded to $15,000+ by the 2020s.
2. **Education** is the biggest differentiator. 78% tertiary enrollment vs 35%.
3. **R&D spending** differs by nearly 3x (1.7% vs 0.6% of GDP). Every single high-R&D observation in our data belongs to a success country.
4. **Governance** is the most consistent gap. About 1 point higher on corruption control, rule of law, and regulatory quality across all four decades.
5. Trapped countries faced far greater macro instability (Brazil's inflation hit 2,948%).
6. Our logistic regression classifies success vs trapped at **92% accuracy**.

The divergence over time

	1980s	1990s	2000s	2010s	2020s
Success	$2,501	$7,120	$12,961	$18,938	$22,834
Trapped	$1,781	$3,384	$4,835	$7,069	$8,185



Korea started at roughly the same level as Thailand. By 2023 it's at 5x.
The gap went from $700 to $15,000.

What drives the gap?

Structural differences:

- Education: 78% vs 35%
- R&D: 1.7% vs 0.6% of GDP
- Patents: 28k vs 1.6k / year
- Governance: +0.8 to +1.1 (all dimensions)

Interaction: Education works best with manufacturing.
Success: 18% have both | Trapped: 1%

Macro instability:

- Brazil hyperinflation: up to 2,948%
- Kept real events; used log scale
- IQR: 47 extreme inflation years (kept)

When we split correlations by group, education correlates much more strongly with GDP in the success group.

Feature engineering: method

We created 12 new features. Example:

```
df['growth_5yr_avg'] = (
    df.groupby('country')['gdp_growth']
    .transform(lambda x: x.rolling(5, min_periods=3).mean())
)
```

`groupby` → `transform` → `lambda` → `rolling mean`. Returns same-index Series for direct column assignment.

Other features: manufacturing intensity ratio, macro instability composite, high-edu × high-manufacturing interaction, income classification + OHE dummies. Scaling: `log1p()` on inflation, per-country interpolation, MinMaxScaler to 0–1.

Feature engineering: what we found

Engineered Feature	Success	Trapped	Takeaway
Macro instability index	13.9	57.2	Trapped 4x worse
Transition readiness score	0.554	0.452	Gap widens over time
High-edu + high-manuf	18%	1%	Almost exclusive to Success

Macro instability combines inflation and unemployment. The readiness score weights 6 scaled indicators and confirms the divergence in raw GDP. Education alone isn't enough, you need it combined with strong manufacturing.

SQL analysis

We loaded the data into 4 normalized tables in SQLite and ran 12 queries covering joins, window functions (RANK, LAG), GROUP BY with ROLLUP/CUBE, and subqueries.

```
WITH rd_stats AS (
  SELECT AVG(rd_pct_gdp) AS mean_rd
  FROM innovation
  WHERE rd_pct_gdp IS NOT NULL
)
SELECT
  CASE WHEN i.rd_pct_gdp >= s.mean_rd
       THEN 'High R&D'
       ELSE 'Low R&D' END AS rd_group,
  c.group_label,
  ROUND(AVG(e.gdp_growth),2)
FROM innovation i
JOIN econ_indicators e
  ON i.country=e.country
  AND i.year=e.year
JOIN countries c ON i.country=c.country
CROSS JOIN rd_stats s
GROUP BY rd_group, c.group_label
```

Result: every single high-R&D observation belongs to a success country. Zero exceptions.

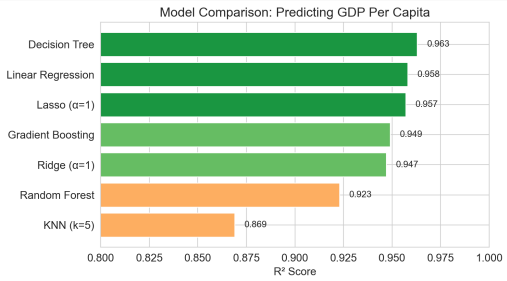
- Brazil, S. Africa, Thailand have never crossed the $13,845 high-income line
- R&D 5-year moving average shows Korea trending to 4.5%+; trapped countries flat at ∼0.5%

Models

Predicting GDP per capita

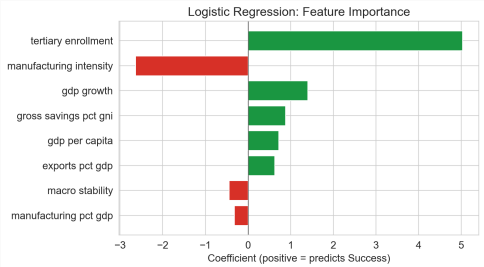
Raw features: $R^2 = 0.57 \rightarrow$ with engineered: $\mathbf{R^2 = 0.96}$

7 models compared:



Top predictor: R&D (74%), then education (17%).

Classifying success vs trapped



Accuracy: **92%** — AUC: **0.97**

Tertiary enrollment strongest positive; manufacturing intensity strongest negative as trapped countries have *higher* manufacturing but lack education and R&D.

So what should policymakers do?

Based on everything we found, three things matter most:

1. **Invest heavily in education and R&D.** These are the top predictors in every model we ran. R&D alone accounts for 74% of feature importance.
2. **Keep inflation and unemployment under control.** You can't build long-term growth on a volatile macro foundation.
3. **Build stronger institutions.** Governance quality is the most *consistent* gap — it shows up in every decade and every metric.

If we had more time: expand to 20+ countries, add UN Comtrade trade data, run panel regressions for causal inference, and look at demographics.